

Finance Economics

Self-Study Lecture Note

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Note: This lecture note was created for the purpose of self-study, based on the content of Professor Cheol Beom Park's Finance Economics class. Redistribution is strictly prohibited as it contains portions directly taken from the lecture notes of Professor Park. Please feel free to report errors or typos via e-mail.

*** Notation may vary: e.g., the CRRA risk aversion index is sometimes written r and other times γ . Interpret from context. ***

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Utility Functions

Quadratic Utility

$$u(c_t) = \alpha + \beta c_t + \gamma c_t^2, \quad u'(c_t) = \beta + 2\gamma c_t$$

Since marginal utility is linear in c_t , many problems admit clean closed-form solutions. Coefficient restrictions require $\beta > 0$ and $\gamma < 0$. The bliss point (where $u' = 0$) is

$$c_t^* = -\frac{\beta}{2\gamma}.$$

Consumption beyond c_t^* reduces utility, so the relevant domain is $c_t < c_t^*$.

Constant Absolute Risk Aversion (CARA)

$$u(c_t) = -e^{-\gamma c_t}, \quad \gamma > 0$$

Here γ is the absolute risk aversion coefficient. The Arrow–Pratt measure is

$$ARA = -\frac{u''(c_t)}{u'(c_t)} = \gamma, \quad u'(c_t) = \gamma e^{-\gamma c_t}, \quad u''(c_t) = -\gamma^2 e^{-\gamma c_t}.$$

Remark: CARA and the MRS

Suppose an individual faces two states with probabilities π_1, π_2 and consumes c_1, c_2 :

$$\pi_1 u(c_1) + \pi_2 u(c_2) = \bar{u}.$$

The MRS along an indifference curve satisfies

$$MRS = -\frac{dc_2}{dc_1} = \frac{\pi_1 u'(c_1)}{\pi_2 u'(c_2)},$$

so

$$\ln(MRS) = \ln \pi_1 + \ln u'(c_1) - \ln \pi_2 - \ln u'(c_2).$$

Differentiating (with $dc_1 = dc_2 = dx$ along the line $c_1 = c_2$):

$$d \ln(MRS) = \left[\frac{u''(c_1)}{u'(c_1)} - \frac{u''(c_2)}{u'(c_2)} \right] dx = [-\gamma - (-\gamma)] dx = 0.$$

Hence an individual's MRS is unchanged by proportional wealth shifts — the hallmark of CARA.

Constant Relative Risk Aversion (CRRA)

$$u(c_t) = \begin{cases} \frac{c_t^{1-\gamma}}{1-\gamma} & \gamma \neq 1, \\ \ln c_t & \gamma = 1. \end{cases}$$

The logarithmic case is the $\gamma \rightarrow 1$ limit, which can be verified by L'Hôpital's theorem:

$$\lim_{\gamma \rightarrow 1} \frac{c_t^{1-\gamma} - 1}{1-\gamma} = \ln c_t.$$

Markowitz Model

Mean-Variance Utility

Assume N risky assets with returns $\tilde{r}_i$. Instead of maximising utility over consumption states directly, we can work with the indirect utility over $E(r)$ and σ^2 :

$$u(E(r_i), \sigma^2).$$

Derivation: Quadratic utility $\Rightarrow$ mean-variance

With initial wealth Y_0 and portfolio return $\tilde{r}$, end-of-period wealth is $Y_0(1 + \tilde{r})$. Under quadratic utility:

$$U^*(Y_0(1 + \tilde{r})) = \alpha + \beta Y_0(1 + \tilde{r}) + \gamma Y_0^2(1 + \tilde{r})^2.$$

Taking expectations:

$$\begin{aligned} \max E[U^*] &= \max E[\alpha + \beta Y_0(1 + \tilde{r}) + \gamma Y_0^2(1 + \tilde{r})^2] \\ &= \max [(\alpha + \beta Y_0 + \gamma Y_0^2) + (\beta Y_0 + 2\gamma Y_0^2) E(\tilde{r}) + \gamma Y_0^2 E(\tilde{r}^2)]. \end{aligned}$$

Using $E(\tilde{r}^2) = \text{Var}(\tilde{r}) + (E(\tilde{r}))^2$:

$$\max E[U^*] = \underbrace{\text{const.}}_{\alpha + \beta Y_0 + \gamma Y_0^2} + (\beta Y_0 + 2\gamma Y_0^2) E(\tilde{r}) + \gamma Y_0^2 [\text{Var}(\tilde{r}) + (E(\tilde{r}))^2].$$

Hence the optimal choice depends only on $E(\tilde{r})$ and $\text{Var}(\tilde{r})$. $\square$

Optimal Weighted Portfolio

We minimise portfolio variance subject to a target return R_p and full investment:

$$\begin{aligned} \min_W \quad & \frac{1}{2} W^\top \Sigma W \\ \text{s.t.} \quad & \bar{R}^\top W = R_p, \quad \iota^\top W = 1, \end{aligned}$$

where $W = (w_1, \dots, w_N)^\top$, $\bar{R} = (E(r_1), \dots, E(r_N))^\top$, $\iota = (1, \dots, 1)^\top$, Σ is the $N \times N$ covariance matrix.

The portfolio variance expands as

$$W^\top \Sigma W = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij} = w_1^2 \sigma_{11} + 2w_1 w_2 \sigma_{12} + \dots + w_N^2 \sigma_{NN}.$$

Derivation: Lagrangian solution

$$\mathcal{L} = \frac{1}{2} W^\top \Sigma W + \lambda_1 (\bar{R}_p - \bar{R}^\top W) + \lambda_2 (1 - \iota^\top W).$$

FOC: $\Sigma W = \lambda_1 \bar{R} + \lambda_2 \iota$, so $W^* = \lambda_1 \Sigma^{-1} \bar{R} + \lambda_2 \Sigma^{-1} \iota$.

Define the scalar constants:

$$A = \bar{R}^\top \Sigma^{-1} \bar{R}, \quad B = \bar{R}^\top \Sigma^{-1} \iota, \quad C = \iota^\top \Sigma^{-1} \iota, \quad D = AC - B^2 > 0.$$

The two constraints give

$$R_p = \lambda_1 A + \lambda_2 B, \quad (1)$$

$$1 = \lambda_1 B + \lambda_2 C. \quad (2)$$

Solving: from (2), $\lambda_2 = (1 - \lambda_1 B)/C$. Substituting into (1):

$$\lambda_1 = \frac{R_p C - B}{D}, \quad \lambda_2 = \frac{A - R_p B}{D}.$$

Minimised variance:

$$\begin{aligned} \sigma_p^2 &= W^{*\top} \Sigma W^* = \lambda_1 R_p + \lambda_2 \cdot 1 \\ &= \frac{(R_p C - B)R_p + (A - R_p B)}{D} = \frac{C R_p^2 - 2B R_p + A}{D}. \end{aligned}$$

Key result — marginal risk of return:

$$\frac{d\sigma_p^2}{dR_p} = \frac{2(CR_p - B)}{D} = \frac{2\lambda_1}{C}.$$

This is positive iff $R_p > B/C = R_G$ (the global minimum variance return), which is the upper branch of the mean-variance frontier.

Global Minimum Variance Portfolio

Dropping the return constraint (only $\iota^\top w = 1$) yields:

$$w_G = \frac{\Sigma^{-1} \iota}{C}, \quad R_G = \bar{R}^\top w_G = \frac{B}{C}.$$

Two-Fund Separation

Any minimum-variance portfolio is a weighted combination:

$$W^* = \underbrace{\lambda_1 B}_{\frac{\lambda_1 B}{\iota^\top \Sigma^{-1} \bar{R}}} \frac{\Sigma^{-1} \bar{R}}{\iota^\top \Sigma^{-1} \bar{R}} + \underbrace{\lambda_2 C}_{\lambda_2 C} w_G^*, \quad \lambda_1 B + \lambda_2 C = 1.$$

Portfolio with a Risk-Free Asset

Allowing investment $w_0 = 1 - \iota^\top W$ in the risk-free asset r_f :

$$\bar{R}_p = \sum_{i=1}^N w_i E(r_i) + \left(1 - \sum_{i=1}^N w_i\right) r_f = r_f + (\bar{R} - r_f \iota)^\top W.$$

Derivation: Optimal portfolio with risk-free asset

$$\min_W \frac{1}{2} W^\top \Sigma W \quad \text{s.t.} \quad \bar{R}_p - r_f = (\bar{R} - r_f \iota)^\top W.$$

Lagrangian:

$$\mathcal{L} = \frac{1}{2} W^\top \Sigma W + \lambda \left[\bar{R}_p - r_f - (\bar{R} - r_f \iota)^\top W \right].$$

FOC: $\Sigma W^* = \lambda(\bar{R} - r_f \iota)$, so

$$W^* = \lambda \Sigma^{-1} (\bar{R} - r_f \iota).$$

Let $E = (\bar{R} - r_f)^\top \Sigma^{-1} (\bar{R} - r_f)$. Then the return constraint gives

$$\bar{R}_p - r_f = \lambda E \implies \lambda = \frac{\bar{R}_p - r_f}{E}.$$

Minimised portfolio risk:

$$\sigma_p^2 = W^{*\top} \Sigma W^* = \lambda^2 E \implies \sigma_p = \lambda \sqrt{E} = \frac{\bar{R}_p - r_f}{\sqrt{E}}.$$

Capital Market Line: $\bar{R}_p - r_f = \sigma_p \sqrt{E}$

This is the equation of the Capital Market Line (CML) — a straight line in $(\sigma_p, \bar{R}_p)$ space emanating from $(0, r_f)$.

Capital Asset Pricing Model (CAPM)

CAPM Assumptions

1. Individuals are price-takers.
2. Single-period model.
3. Risk-free return r_f is given.
4. No transaction costs or taxes.
5. Mean-variance utility (Markowitz framework).
6. **Homogeneous expectations:** every investor shares the same beliefs about returns, variances, and covariances.

Each investor solves

$$\max_{w_i} U \left(\sum_{i=1}^N w_i E(r_i) + \left(1 - \sum_{i=1}^N w_i \right) r_f, \sum_{i,j} w_i w_j \sigma_{ij} \right).$$

FOC for asset i :

$$\frac{\partial U}{\partial w_i} = 0 \Leftrightarrow u_1 (E(r_i) - r_f) + 2u_2 \sum_{j=1}^N w_j^* \sigma_{ij} = 0,$$

where $u_1 = \partial U / \partial E(r_c)$ and $u_2 = \partial U / \partial \sigma_c^2$.

Derivation: Interpreting $\sum_j w_j^* \sigma_{ij}$

$$\begin{aligned} \sum_{j=1}^N w_j^* \sigma_{ij} &= \sum_{j=1}^N w_j^* E[(r_i - E(r_i))(r_j - E(r_j))] \\ &= E \left[(r_i - E(r_i)) \sum_{j=1}^N w_j^* (r_j - E(r_j)) \right] \\ &= \text{Cov}(r_i, r_m), \end{aligned}$$

since the risk-free return is constant and drops out of the covariance.

Substituting $\sum_j w_j^* \sigma_{ij} = \text{Cov}(r_i, r_m)$ back into the FOC gives, for each asset i :

$$\begin{aligned} E(r_1) - r_f &= -\frac{2u_2}{u_1} \sum_{j=1}^N w_j^* \sigma_{1j} \\ E(r_2) - r_f &= -\frac{2u_2}{u_1} \sum_{j=1}^N w_j^* \sigma_{2j} \\ &\vdots \\ E(r_N) - r_f &= -\frac{2u_2}{u_1} \sum_{j=1}^N w_j^* \sigma_{Nj} \end{aligned}$$

Multiply the i -th equation by w_i^* and sum over all i :

$$\begin{aligned} \sum_{i=1}^N w_i^* [E(r_i) - r_f] &= -\frac{2u_2}{u_1} \sum_{i=1}^N \sum_{j=1}^N w_i^* w_j^* \sigma_{ij} \\ \Rightarrow \sum_{i=1}^N w_i^* E(r_i) + \left(1 - \sum_{i=1}^N w_i^*\right) r_f - r_f &= -\frac{2u_2}{u_1} \sigma_m^2 \\ \therefore E(r_m) - r_f &= -\frac{2u_2}{u_1} \sigma_m^2 \Rightarrow -\frac{2u_2}{u_1} = \frac{E(r_m) - r_f}{\sigma_m^2}. \end{aligned}$$

CAPM Security Market Line:

$$E(r_i) - r_f = \beta_i (E(r_m) - r_f), \quad \beta_i = \frac{\text{Cov}(r_i, r_m)}{\sigma_m^2}.$$

Example: Covariance and equilibrium price

Stock	P_0	Eq. Price	$E(P_1)$	Covariance
A	\$10,000	Increases	\$12,000	$\text{Cov}(r_A, r_m) < 0$
B	\$10,000	Decreases	\$12,000	$\text{Cov}(r_B, r_m) > 0$

Stock B has positive covariance with the market — it falls in bad times — so investors demand a higher expected return, driving its equilibrium price down.

Empirical Test: Fama & MacBeth (1973, JF)

The CAPM predicts $E(r_i) - r_f = \beta_i (E(r_m) - r_f)$, i.e. the intercept in

$$E(r_i) - r_f = \alpha_i + \beta_i (E(r_m) - r_f) + \varepsilon_i$$

should be zero.

1. Time-series regression (estimate $\hat{\beta}_i$):

$$r_{it} - r_{ft} = \alpha_i + \beta_i (r_{mt} - r_{ft}) + \varepsilon_{it}.$$

2. Cross-sectional regression (each period t):

$$R_{it}^e = \lambda_t \hat{\beta}_i + \alpha_t + u_i.$$

3. t-test on $\bar{\alpha}$:

$$\bar{\alpha} = \frac{1}{T} \sum_{t=1}^T \hat{\alpha}_t, \quad \hat{\sigma}_\alpha^2 = \frac{1}{T(T-1)} \sum_{t=1}^T (\hat{\alpha}_t - \bar{\alpha})^2, \quad t = \frac{\bar{\alpha}}{\hat{\sigma}_\alpha} \sim t(T-1).$$

Single Factor Model

The mean-variance approach requires $N + \frac{N(N+1)}{2}$ estimates (means + covariances), creating over-fitting risk. The single-factor model reduces this.

$$r_i = \alpha_i + \beta_i M + \varepsilon_i$$

Term	Meaning
r_i	Return of asset i
α_i	Asset-specific constant (company characteristics, time-invariant)
M	Common macro factor (e.g. market return in CAPM)
ε_i	Idiosyncratic risk, $E(\varepsilon_i) = 0$, uncorrelated across i

Key simplifications:

$$E(r_i) = \alpha_i + \beta_i E(M) \quad (\text{only } 2N + 1 \text{ estimates}),$$

$$\text{Var}(r_i) = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon_i}^2,$$

$$\text{Cov}(r_i, r_j) = \beta_i \beta_j \sigma_M^2.$$

Total estimates needed: $3N + 2$ (i.e. N α_i 's, N β_i 's, N $\sigma_{\varepsilon_i}^2$'s, plus $E(M)$ and σ_M^2).

Consumption Capital Asset Pricing Model (CCAPM)

CAPM uses the market portfolio as the sole pricing factor; CCAPM links asset prices to the consumer's intertemporal optimisation.

Consumer's problem:

$$\begin{aligned} \max \quad & E_t \left[\sum_{i=0}^{\infty} \left(\frac{1}{1+\rho} \right)^i u(c_{t+i}) \right] \\ \text{s.t.} \quad & c_t + p_t a_{t+1} = e_t + a_t \chi_{t+1}, \end{aligned}$$

Term	Meaning
c_t	Consumption at time t
a_t	Asset holdings entering period t

e_t	Endowment income
χ_{t+1}	Next-period payoff (dividend + resale price)
ρ	Subjective discount rate

For a **risky asset**: payoff $\chi_{t+1} = D_{t+1} + P_{t+1}$, gross return $R_{t+1} = (D_{t+1} + P_{t+1})/P_t$.

For a **risk-free asset**: payoff = 1, price = P_f , gross return $R_f = 1/P_f$.

Euler Equation and the Stochastic Discount Factor

Starting from the optimal consumption plan $\{C_t^*, C_{t+1}^*, C_{t+2}^*, \dots\}$, consider two alternative one-unit deviations:

$$\begin{aligned}
\{C_t^*, C_{t+1}^*, C_{t+2}^*, \dots\} & \quad \text{(optimal)} \\
\{C_t^* - P_t, C_{t+1}^* + \chi_{t+1}, C_{t+2}^*, \dots\} & \quad \text{(buy one unit now)} \\
\{C_t^* + P_t, C_{t+1}^* - \chi_{t+1}, C_{t+2}^*, \dots\} & \quad \text{(borrow against future)}
\end{aligned}$$

Since the optimal plan maximises utility, both deviations (2) and (3) yield weakly lower utility than (1). This gives two inequalities; combining them (both ≤ 0) yields the **Euler equation**:

$$u'(C_t^*) P_t = E_t \left[\frac{1}{1 + \rho} u'(C_{t+1}^*) \chi_{t+1} \right].$$

Dividing by $u'(C_t^*) P_t$:

$$1 = E_t[m_{t+1} R_{t+1}], \quad m_{t+1} \equiv \frac{1}{1 + \rho} \frac{u'(C_{t+1}^*)}{u'(C_t^*)}, \quad R_{t+1} = \frac{\chi_{t+1}}{P_t}.$$

m_{t+1} is the **Stochastic Discount Factor (SDF)**: it discounts future payoffs by both time preference and the marginal utility of future consumption relative to today.

Asset price as discounted sum of dividends:

$$\begin{aligned}
P_t &= E_t[m_{t+1} \chi_{t+1}] = E_t[m_{t+1} d_{t+1}] + E_t[m_{t+1} P_{t+1}] \\
&= E_t[m_{t+1} d_{t+1}] + E_t[m_{t+1} (E_{t+1}[m_{t+2} d_{t+2}] + E_{t+1}[m_{t+2} P_{t+2}])] \\
&= E_t[m_{t,t+1} d_{t+1}] + E_t[m_{t,t+2} d_{t+2}] + E_t[m_{t,t+3} d_{t+3}] + \dots
\end{aligned}$$

where $m_{t,t+j} = \left(\frac{1}{1+\rho} \right)^j \frac{u'(C_{t+j}^*)}{u'(C_t^*)}$, obtained by telescoping products of one-period SDFs.

Under the **no-bubble condition**:

$$\lim_{j \rightarrow \infty} E_t[m_{t,t+j} P_{t+j}] = 0, \quad \therefore P_t = \sum_{j=1}^{\infty} E_t[m_{t,t+j} d_{t+j}].$$

Euler Equation: Risk-Free Asset

$$1 = E_t \left[\frac{m_{t+1}}{P_f} \right] = E_t[m_{t+1}] R_f \implies R_f = \frac{1}{E_t[m_{t+1}]}.$$

CRRA case ($u(C) = C^{1-\gamma}/(1-\gamma)$, $\beta = 1/(1+\rho) = e^{-\delta}$):

$$m_{t+1} = e^{-\delta} \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} = e^{-\delta - \gamma \Delta \ln C_{t+1}}.$$

Using $\ln E[e^X] = E[X] + \frac{1}{2} \text{Var}(X)$ (log-normality):

$$r_f \equiv \ln R_f = \delta + \gamma E_t[\Delta \ln C_{t+1}] - \frac{\gamma^2}{2} \sigma_t^2(\Delta \ln C_{t+1}).$$

Interpretation:

- Higher expected consumption growth ($E_t[\Delta \ln C_{t+1}] \uparrow \Rightarrow r_f \uparrow$ (households save less, pushing up the interest rate).
- Higher consumption volatility ($\sigma_t^2 \uparrow \Rightarrow r_f \downarrow$ (precautionary saving pushes down the interest rate)).

Euler Equation: Risky Asset

$$P_t = E_t[m_{t+1}] E_t[\chi_{t+1}] + \text{Cov}_t(m_{t+1}, \chi_{t+1}) = \frac{E_t[\chi_{t+1}]}{R_f} + \text{Cov}_t(m_{t+1}, \chi_{t+1}).$$

Economic interpretation

$$\underbrace{\frac{E_t[\chi_{t+1}]}{R_f}}_{\text{risk-neutral value}} + \underbrace{\text{Cov}_t(m_{t+1}, \chi_{t+1})}_{\text{risk-adjustment}}.$$

If $\text{Cov}_t(m_{t+1}, \chi_{t+1}) < 0$, the asset pays off poorly when marginal utility is high (bad times) $\Rightarrow$ investors require a discount (lower price / higher return).

Expressed as an **excess return**:

$$E_t[R_{t+1}] - R_f = - \frac{\text{Cov}_t(m_{t+1}, R_{t+1})}{E_t[m_{t+1}]}.$$

Idiosyncratic Risk

Suppose a new asset pays $\hat{\chi}_{t+1} = \chi_{t+1} + e_{t+1}$ where $e_{t+1} \perp m_{t+1}$, $E_t(e_{t+1}) = 0$.

$$\hat{P}_t = \frac{E_t(\chi_{t+1}) + E_t(e_{t+1})}{R_f} + \text{Cov}_t(m_{t+1}, \chi_{t+1}) + \underbrace{\text{Cov}_t(m_{t+1}, e_{t+1})}_{=0} = P_t.$$

Key Insight

Idiosyncratic risk (e_{t+1}) that is uncorrelated with the SDF has **no effect on price**. Only **systematic risk** (covariance with m_{t+1}) is priced.

Mean-Variance Frontier

Rewrite the excess return equation using $\text{Cov} = \text{Corr} \cdot \sigma \cdot \sigma$:

$$E_t(R_{t+1}^i) - R_f = - \frac{\text{Cov}_t(m_{t+1}, R_{t+1}^i)}{E_t(m_{t+1})} = - \frac{\text{Corr}_t(m_{t+1}, R_{t+1}^i)}{E_t(m_{t+1})} \sigma_t(m_{t+1}) \sigma_t(R_{t+1}^i).$$

Since $|\text{Corr}| \leq 1$, the risk premium is bounded:

Case 1: $\text{Corr}_t = -1$ (upper frontier):

$$E_t(R_{t+1}^i) - R_f = + \frac{\sigma_t(m_{t+1})}{E_t(m_{t+1})} \sigma_t(R_{t+1}^i).$$

Case 2: $\text{Corr}_t = +1$ (lower frontier):

$$E_t(R_{t+1}^i) - R_f = - \frac{\sigma_t(m_{t+1})}{E_t(m_{t+1})} \sigma_t(R_{t+1}^i).$$

These two lines form the **Mean-Variance Frontier** in $(\sigma_t(R^i), E_t(R^i))$ space.

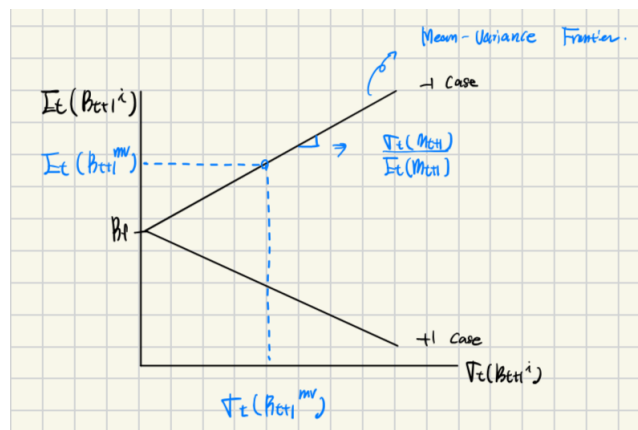


Figure 1: Mean-Variance Frontier

Any portfolio on the upper frontier satisfies $\text{Corr}_t(m_{t+1}, R_{t+1}^{mv}) = -1$, so for any asset i :

$$\begin{aligned} E_t(R_{t+1}^i) - R_f &= - \frac{\text{Cov}_t(m_{t+1}, R_{t+1}^i)}{E_t(m_{t+1})} \\ &= - \frac{\text{Corr}_t(R_{t+1}^{mv}, R_{t+1}^i) \sigma_t(m_{t+1}) \sigma_t(R_{t+1}^i)}{E_t(m_{t+1})} \\ &= \beta_{i,mv} [E_t(R_{t+1}^{mv}) - R_f]. \end{aligned}$$

CCAPM vs CAPM

In CAPM the pricing factor is the **market portfolio** return. In CCAPM the pricing factor is the **SDF** — ultimately **consumption growth**.

Remark: CRRA utility and the SDF

Under CRRA ($u(C_t) = \frac{C_t^{1-\gamma}}{1-\gamma}$, so $u'(C_t) = C_t^{-\gamma}$):

$$m_{t+1} = \frac{1}{1+\rho} \frac{u'(C_{t+1})}{u'(C_t)} = \frac{1}{1+\rho} \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma}.$$

γ controls both risk aversion and the sensitivity of the SDF to consumption growth.

CRRRA approximation ($C_{t+1}/C_t \approx 1 + g_{t+1}$, so $(1 + g)^{-\gamma} \approx 1 - \gamma g$):

$$E_t(R_{t+1}^i) - R_f \approx \frac{R_f \gamma}{1 + \rho} \text{Cov}_t(g_{t+1}, R_{t+1}^i).$$

Example: Covariance and price — CCAPM

Stock	P_0	Eq. Price	$E(P_1)$	Covariance
A	\$10,000	Increases	\$12,000	$\text{Cov}(g_{t+1}, R_{t+1}^A) < 0$
B	\$10,000	Decreases	\$12,000	$\text{Cov}(g_{t+1}, R_{t+1}^B) > 0$

Asset A pays well when consumption is low (insurance) $\Rightarrow$ high demand $\Rightarrow$ high price.
 Asset B falls in recessions $\Rightarrow$ requires a risk premium $\Rightarrow$ lower price.

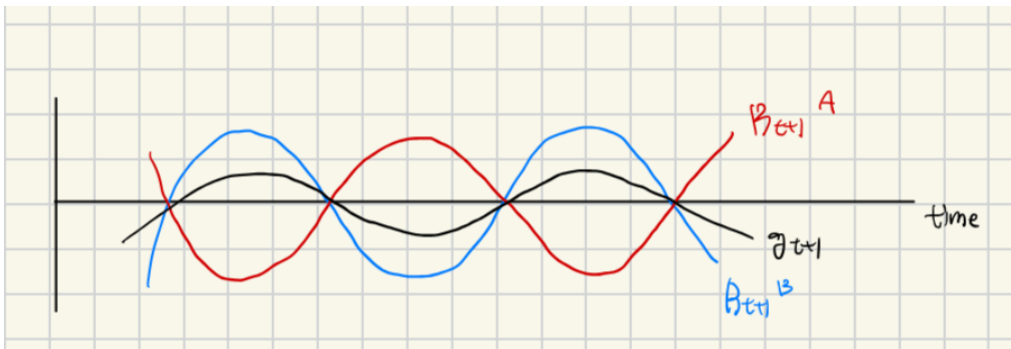


Figure 2: Covariance between consumption growth and asset return

Is There a Predictable Component in Asset Prices?

Recall the Euler equation:

$$u'(C_t^*)P_t = E_t \left[\frac{1}{1 + \rho} u'(C_{t+1}^*) (d_{t+1} + p_{t+1}) \right].$$

At **high frequency** (short time intervals), the discount rate ρ and consumption growth are negligible, so:

$$P_t \approx E_t[P_{t+1}].$$

Hence the price follows a **random walk**:

$$P_t = P_{t+1} + \varepsilon_{t+1}.$$

Under rational expectations:

$$E_t(\varepsilon_{t+1}) = 0, \quad \text{Cov}(\varepsilon_t, \varepsilon_{t+j}) = 0 \quad \forall j > 0.$$

Since $P_{t+1} - P_t = \mu + \varepsilon_t$ is the return r_{t+1} , we have:

$$\text{Cov}(\varepsilon_t, \varepsilon_{t+j}) = \text{Cov}(r_t, r_{t+j}).$$

So testing whether ε_t is white noise is equivalent to testing whether stock *returns* are serially uncorrelated.

Autocorrelation Test

Testing for White Noise in Asset Returns

We want to check whether ε_t (the shock to the stock price) is white noise, i.e. has zero autocorrelations:

$$\text{Cov}(r_t, r_{t-j}) = E[(r_t - E(r_t))(r_{t-j} - E(r_{t-j}))].$$

Under stationarity ($E(r_t) = E(r_{t-j})$), the sample autocovariance is:

$$\hat{\gamma}_{(k)} = \frac{1}{T} \sum_{t=1}^{T-k} (r_t - \bar{r})(r_{t+k} - \bar{r}), \quad \bar{r} = \frac{1}{T} \sum_{t=1}^T r_t.$$

And the **sample autocorrelation at lag k** :

$$\hat{\rho}_{(k)} = \frac{\hat{\gamma}_{(k)}}{\hat{\gamma}_{(0)}}.$$

Box–Ljung (Portmanteau) Test

If $\varepsilon_t \sim MA(q)$, asymptotic variances of the sample autocorrelations are:

$$\text{Var}(\hat{\rho}_{(1)}) = \frac{1}{T}, \quad \text{Var}(\hat{\rho}_{(k)}) = \frac{1}{T} \left[1 + 2 \sum_{j=1}^{k-1} \rho_j^2 \right].$$

Asymptotically, $\sqrt{T}[\hat{\rho}_{(k)} - \rho_{(k)}] \sim N(0, G)$ and $\hat{\rho}_{(k)} \perp \hat{\rho}_{(j)}$ for $k \neq j$.

$$H_0: P_t \sim \text{Random Walk} \Leftrightarrow \rho_{(k)} = 0 \text{ for all } k > 0.$$

Variance Ratio Test, Hausman (1978)

Key identity:

$$\text{Var}(r_t + r_{t+1}) = \text{Var}(r_t) + \text{Var}(r_{t+1}) + 2\text{Cov}(r_t, r_{t+1}) = 2\text{Var}(r_t) + 2\text{Cov}(r_t, r_{t+1}).$$

$$VR(2) = \frac{\text{Var}(r_t + r_{t+1})}{2\text{Var}(r_t)} = 1 + \rho_{(1)}.$$

Under Random Walk: $H_0 : VR(2) = 1$ because $\rho_{(1)} = 0$.

Three-period extension:

$$VR(3) = 1 + \frac{4}{3}\rho_{(1)} + \frac{2}{3}\rho_{(2)}.$$

General formula:

$$VR(q) = \frac{\text{Var}(r_t + \dots + r_{t+q-1})}{q\text{Var}(r_t)} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_{(k)}.$$

Under RW: $H_0 : VR(q) = 1$.

Asymptotic Distribution of VR(2)

With $2n$ returns $p_0, \dots, p_{2n}$, define

$$\hat{\mu} = \frac{p_{2n} - p_0}{2n}, \quad \hat{\sigma}_a^2 = \frac{1}{2n} \sum_{k=1}^{2n} (p_k - p_{k-1} - \hat{\mu})^2, \quad \hat{\sigma}_b^2 = \frac{1}{2n} \sum_{k=1}^n (p_{2k} - p_{2k-2} - 2\hat{\mu})^2.$$

Then $VR(2) = \hat{\sigma}_b^2 / \hat{\sigma}_a^2$. Asymptotically:

$$\sqrt{2n}(\hat{\sigma}_a^2 - \sigma^2) \sim N(0, 2\sigma^4), \quad \sqrt{2n}(\hat{\sigma}_b^2 - \sigma^2) \sim N(0, 4\sigma^4).$$

Hausman (1978) shows $\hat{\sigma}_a^2 \perp (\hat{\sigma}_a^2 - \hat{\sigma}_b^2)$ asymptotically. The joint distribution is:

$$\sqrt{2n} \begin{pmatrix} \hat{\sigma}_a^2 - \hat{\sigma}_b^2 \\ \hat{\sigma}_a^2 - \sigma^2 \end{pmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2\sigma^4 & 0 \\ 0 & 2\sigma^4 \end{bmatrix} \right).$$

Delta Method

If $Y_n \sim N(\theta, \sigma^2/n)$ and g is differentiable at θ :

$$g(Y_n) \approx N \left(g(\theta), [g'(\theta)]^2 \frac{\sigma^2}{n} \right).$$

Matrix form: if $\sqrt{T}(\hat{\theta} - \theta_0) \sim N(0, V_\theta)$, then

$$\sqrt{T}(f(\hat{\theta}) - f(\theta_0)) \sim N \left(0, \frac{\partial f'}{\partial \theta} V_\theta \frac{\partial f}{\partial \theta} \right).$$

Applying the Delta Method to $f(\hat{\theta}_1, \hat{\theta}_2) = \hat{\theta}_1 / \hat{\theta}_2 = (\hat{\sigma}_a^2 - \hat{\sigma}_b^2) / \hat{\sigma}_a^2$:

$$\frac{\partial f}{\partial \theta_1} = \frac{1}{\theta_2} = \frac{1}{\sigma^2}, \quad \frac{\partial f}{\partial \theta_2} = -\frac{\theta_1}{\theta_2^2} \rightarrow 0 \quad (\text{under } H_0 : \theta_1 = 0).$$

$$\begin{pmatrix} \frac{1}{\sigma^2} & 0 \end{pmatrix} \begin{pmatrix} 2\sigma^4 & 0 \\ 0 & 2\sigma^4 \end{pmatrix} \begin{pmatrix} \frac{1}{\sigma^2} \\ 0 \end{pmatrix} = 2.$$

$$\sqrt{2n}(\widehat{VR}(2) - 1) = \sqrt{2n} \frac{\hat{\sigma}_b^2 - \hat{\sigma}_a^2}{\hat{\sigma}_a^2} \xrightarrow{d} N(0, 2).$$

Lo & MacKinlay (1988)

Hausman used non-overlapping two-period returns, wasting half the data. Lo & MacKinlay use **overlapping** windows:

$$r_1 + r_2, r_2 + r_3, \dots, r_{n-1} + r_n.$$

With ng observations and holding period g :

$$\hat{\mu} = \frac{p_{ng} - p_0}{ng}, \quad \hat{\sigma}_a^2 = \frac{1}{ng-1} \sum_{k=1}^{ng} (p_k - p_{k-1} - \hat{\mu})^2,$$

$$\hat{\sigma}_c^2 = \frac{1}{m} \sum_{k=1}^{ng} (p_k - p_{k-g} - g\hat{\mu})^2, \quad m = g(ng - g + 1) \left(1 - \frac{1}{n}\right).$$

$$\sqrt{ng}(\widehat{VR}(g) - 1) \sim N \left(0, \frac{2(2g-1)(g-1)}{3g} \right).$$

Unit Root Test

The random walk hypothesis fixes the AR coefficient at 1. We now test whether it actually equals 1.

$$p_t = \mu + \rho p_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2). \quad H_0 : \rho = 1.$$

Derivation: Why does $\rho = 1$ matter? (impulse response)

Using the lag operator L (so $Ly_t = y_{t-1}$):

$$\begin{aligned} y_t &= \rho y_{t-1} + u_t \\ (1 - \rho L)y_t &= u_t \\ y_t &= \frac{u_t}{1 - \rho L} = u_t + \rho u_{t-1} + \rho^2 u_{t-2} + \dots \end{aligned}$$

The impulse response of y_t to a shock at $t - i$ is:

$$\frac{\partial y_t}{\partial u_{t-i}} = \frac{\partial y_{t+i}}{\partial u_t} = \rho^i.$$

- If $|\rho| < 1$: $\rho^i \rightarrow 0$ — shocks die out (**short-run impact**).
- If $\rho = 1$: $\rho^i = 1$ for all i — shocks are **permanent**.

When $\rho = 1$: $y_t = y_0 + (u_1 + u_2 + \dots + u_t)$, so a demand shock at $t = 0$ shifts every future y_t by 1. This models permanent supply shocks in economics.

Law of One Price and Purchasing Power Parity

Absolute PPP: $E_t \cdot P_t^* = P_t$.

Relative PPP: $E_t \cdot P_t^* = K \cdot P_t$, or in logs:

$$\underbrace{e_t + p_t^* - p_t}_{\equiv g_t} = k.$$

PPP holds iff g_t is **stationary** (mean-reverting around k).

Augmented Dickey–Fuller (ADF) Test

Starting from an $AR(p)$ model for g_t and reparametrising:

$$g_t = c + \rho g_{t-1} + r_2 \Delta g_{t-2} + r_3 \Delta g_{t-3} + \dots + r_{p-1} \Delta g_{t-p+1} + \varepsilon_t.$$

Case 1: No constant, no trend ($c = 0$, pure random walk).

Case 2: Constant only ($c \neq 0$, random walk around a mean) — **used for PPP**.

Case 3: Constant + drift ($c \neq 0$, trending mean).

Case 4: Constant + deterministic trend ($c \neq 0$, δt added).

Testing PPP (Case 2):

$$g_t = c + \rho g_{t-1} + r_2 \Delta g_{t-2} + \dots + \varepsilon_t, \quad t = \frac{\hat{\rho} - 1}{\hat{\sigma}_\rho}.$$

Fail to reject $H_0 : \rho = 1 \Rightarrow$ PPP does not hold.

KPSS Test

Lothian & Taylor (1997) note that ADF has low power in small samples when ρ is close to 1. The KPSS test reverses the hypotheses:

$$\text{ADF: } H_0 : \rho = 1 \text{ vs. } H_1 : \rho < 1 \text{ (PPP).} \quad \text{KPSS: } \underbrace{H_0 : \rho < 1}_{\text{PPP}} \text{ vs. } H_1 : \rho = 1.$$

Test statistic:

$$S = \frac{\hat{\varepsilon}^\top V \hat{\varepsilon}}{T^2 \hat{\sigma}_\varepsilon^2}, \quad V_{ij} = \min(i, j),$$

$$\hat{\varepsilon}^\top V \hat{\varepsilon} = \varepsilon_1^2 + (\varepsilon_1 + \varepsilon_2)^2 + \dots + (\varepsilon_1 + \dots + \varepsilon_T)^2.$$

Long-run variance estimate (Bartlett kernel):

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T \hat{\varepsilon}_t^2 + \frac{2}{T} \sum_{l=1}^L \left(1 - \frac{l}{L+1}\right) \sum_{t=l+1}^T \hat{\varepsilon}_t \hat{\varepsilon}_{t-l}, \quad L = \text{int} \left[12 \left(\frac{T}{100} \right)^{1/4} \right].$$

Under H_0 , S converges; under H_1 ($\rho = 1$), S diverges.

Panel Unit Root Test

Combining cross-sectional units to increase power (less common today due to cross-sectional dependence):

$$H_0 : \rho_j = 1 \forall j \quad \text{vs.} \quad H_1 : \rho_j < 1 \forall j.$$

Maddala & Wu (1999) Fisher test:

$$\lambda = -2 \sum_{j=1}^N \ln(p_j) \xrightarrow{d} \chi_{2N}^2,$$

where p_j is the p -value of the ADF test for unit j .

Im, Pesaran & Shin (2003):

$$z = \frac{\sqrt{N}(\bar{t} - E[\bar{t}])}{\sqrt{\text{Var}(\bar{t})}}, \quad \bar{t} = \frac{1}{N} \sum_{j=1}^N t_{\rho_j}.$$

Equity Premium Puzzle

Recall the Euler equation and the excess return formula:

$$E_t(R_{t+1}) - R_f = - \frac{\text{Corr}_t(m_{t+1}, R_{t+1}) \sigma_t(m_{t+1})}{E_t(m_{t+1})} \sigma_t(R_{t+1}).$$

Under the maximal correlation assumption ($\text{Corr}_t = -1$):

$$\frac{E_t(R_{t+1}) - R_f}{\sigma_t(R_{t+1})} = \frac{\sigma_t(m_{t+1})}{E_t(m_{t+1})}.$$

CRRA computation. With $m_{t+1} = \beta e^{-r \Delta \ln C_{t+1}}$:

$$E_t(m_{t+1}) = \beta e^{-r E_t(\Delta \ln C_{t+1}) + \frac{r^2}{2} \sigma_t^2(\Delta \ln C_{t+1})},$$

$$\sigma_t^2(m_{t+1}) = [E_t(m_{t+1})]^2 \left(e^{r^2 \sigma_t^2 (\Delta \ln C_{t+1})} - 1 \right),$$

$$\therefore \frac{\sigma_t(m_{t+1})}{E_t(m_{t+1})} = \sqrt{e^{r^2 \sigma_t^2 (\Delta \ln C_{t+1})} - 1}.$$

First-order Taylor expansion around $\sigma^2 = 0$: let $f(\sigma^2) = e^{r^2 \sigma^2} - 1$, so $f(0) = 0$ and $f'(0) = r^2$. Hence $f(\sigma^2) \approx r^2 \sigma^2$, giving $\sqrt{e^{r^2 \sigma^2} - 1} \approx r \sigma$. Therefore:

$$E_t(R_{t+1}) - R_f \approx r \cdot \sigma_t(\Delta \ln C_{t+1}) \cdot \sigma_t(R_{t+1}).$$

The Puzzle

Empirically: $E(R_{t+1}) - R_f \approx 0.06$, $\sigma(\Delta \ln C_{t+1}) \approx 0.01$, $\sigma(R_{t+1}) \approx 0.16$. This implies $r \approx 38$ — an implausibly high risk aversion coefficient.

Proposed Solutions

Campbell & Cochrane (1999, JPE) — Habit Formation

Modify utility by introducing a “habit” level X_t :

$$u_t(C_t, X_t) = \frac{(C_t - X_t)^{1-r} - 1}{1-r}.$$

First derivative: $u_C = (C_t - X_t)^{-r} = (S_t C_t)^{-r}$, where $S_t = (C_t - X_t)/C_t$ is the **surplus ratio**. SDF:

$$M_{t+1}^c = \beta \left(\frac{S_{t+1} C_{t+1}}{S_t C_t} \right)^{-r}, \quad m_{t+1}^c = \ln \beta - r[\Delta \ln C_{t+1} + \Delta \ln S_{t+1}].$$

Sharpe ratio:

$$\frac{E_t(R_{t+1}) - R_f}{\sigma_t(R_{t+1})} \approx r \cdot \sigma_t(\Delta \ln C_{t+1}) \cdot \sigma_t(h(S_t)).$$

Remark: Economic Intuition

In recessions, $S_t = (C_t - X_t)/C_t$ falls (habit X_t is sticky), making marginal utility very sensitive to consumption. The function $h(S_t)$ amplifies risk aversion in downturns, explaining the high equity premium without requiring $r \approx 38$ unconditionally.

Epstein & Zin (1989) / Kreps & Porteus (1978) — Recursive Utility

CRRA conflates risk aversion and intertemporal substitutability:

$$\phi_{\text{EIS}} = \frac{1}{r} \quad (\text{CRRA only}).$$

Epstein-Zin utility separates them:

$$u_t = \left[(1-\delta) C_t^{\frac{1-r}{\theta}} + \delta (E_t[u_{t+1}^{1-r}])^{\frac{1}{\theta}} \right]^{\frac{\theta}{1-r}}, \quad \theta = \frac{1-r}{1-1/\phi}.$$

Log SDF:

$$m_{t+1} = \theta \ln \delta - \frac{\theta}{\phi} \Delta \ln C_{t+1} - (1-\theta) r_{w,t+1},$$

where $r_{w,t+1}$ is the log return on the wealth portfolio.

Equity premium (log-normal):

$$\begin{aligned} E_t(r_{t+1}) - r_f &= -\frac{1}{2}\sigma_t^2(r_{t+1}) - \text{Cov}_t(m_{t+1}, r_{t+1}) \\ &= -\frac{1}{2}\sigma_t^2(r_{t+1}) + \frac{\theta}{\phi}\text{Cov}_t(\Delta \ln C_{t+1}, r_{t+1}) + (1 - \theta)\text{Cov}_t(r_{w,t+1}, r_{t+1}). \end{aligned}$$

The additional term $(1 - \theta)\text{Cov}_t(r_w, r)$ allows the model to match the equity premium with more plausible r and ϕ .

Luttmer (1999, JPE) — Market Friction

Entering the risky asset market incurs stress a (due to volatility); risk-free market incurs smaller stress $b < a$:

$$p_t u'(C_t - a) = E_t[\beta u'(C_{t+1})(p_{t+1} + d_{t+1})], \quad P_f u'(C_t - b) = E_t[\beta u'(C_{t+1})].$$

This gives:

$$E_t(R_{t+1}) - R_f = \underbrace{\left(\frac{1}{E_t(M^a)} - \frac{1}{E_t(M^b)} \right)}_{>0} + \frac{\beta}{E_t(M^a)} r \sigma_t(R_{t+1}) \sigma_t(\Delta C_{t+1}^a).$$

The positive friction term inflates the equity premium, allowing r to fall from ~ 38 to a plausible 2–3 (with $a \approx 2\text{--}3\%$ of C_t).

Stock Return Predictability

Log-Linearisation of Returns (Campbell & Shiller)

$$r_{t+1} = \ln(D_{t+1} + P_{t+1}) - \ln P_t.$$

Let $dp_t = d_t - p_t$ (log dividend-price ratio), $\bar{dp}$ its mean. Taylor-expand $\ln(1 + e^{dp_{t+1}})$ around $\bar{dp}$, define $\rho = 1/(1 + e^{\bar{dp}})$, $k = -\ln \rho - (1 - \rho) \ln(1/\rho - 1)$. Then:

$$r_{t+1} \approx k + \rho p_{t+1} + (1 - \rho)d_{t+1} - p_t.$$

Iterating forward (no-bubble condition: $\lim_{j \rightarrow \infty} \rho^j p_{t+j} = 0$):

$$p_t = (1 - \rho) \sum_{j=0}^{\infty} \rho^j d_{t+1+j} - \sum_{j=0}^{\infty} \rho^j r_{t+1+j} + \frac{k}{1 - \rho}.$$

Subtracting d_t from both sides:

$$d_t - p_t = - \sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j} + \sum_{j=0}^{\infty} \rho^j r_{t+1+j} - \frac{k}{1 - \rho}.$$

A high price relative to dividends ($d_t - p_t$ low) must be offset by either high future dividend growth or low future returns.

Predictive Regressions

1. $r_{t+1} + \dots + r_{t+k} = \beta_0 + \beta_1(k)(d_t - p_t) + u_{t,t+k}$
2. $\Delta d_{t+1} + \dots + \Delta d_{t+k} = \gamma_0 + \gamma_1(k)(d_t - p_t) + u_{t,t+k}$

Empirically, $H_0 : \gamma_1 = 0$ is rarely rejected, but $H_0 : \beta_1 = 0$ often is — the D/P ratio predicts **returns**, not dividend growth.

Shiller (1981) and Return Variance

Suppose expected returns have a predictable component:

$$E_t(r_{t+1}) = \bar{r} + x_t, \quad x_t = \phi x_{t-1} + \varepsilon_t.$$

Then $P_{r_t} \equiv E_t(\sum_{j=0}^{\infty} \rho^j r_{t+1+j}) = \frac{\bar{r}}{1-\rho} + \frac{x_t}{1-\rho\phi}$.

Derivation: Derivation of the return decomposition

From the Campbell-Shiller approximation $r_{t+1} = k + \rho p_{t+1} - p_t + (1-\rho)d_{t+1}$, substitute:

$$\begin{aligned} p_t &= E_t \left(\sum_{j=0}^{\infty} (1-\rho) \rho^j d_{t+1+j} \right) - E_t \left(\sum_{j=0}^{\infty} \rho^j r_{t+1+j} \right) + \frac{k}{1-\rho}, \\ p_{t+1} &= E_{t+1} \left(\sum_{j=0}^{\infty} (1-\rho) \rho^j d_{t+2+j} \right) - E_{t+1} \left(\sum_{j=0}^{\infty} \rho^j r_{t+2+j} \right) + \frac{k}{1-\rho}. \end{aligned}$$

After cancellation (and adding $\pm d_t$), the *unexpected* return satisfies:

$$\begin{aligned} r_{t+1} - E_t(r_{t+1}) &= \underbrace{\left[E_{t+1} \left(\sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j} \right) - E_t \left(\sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j} \right) \right]}_{\eta_{d,t+1}} \\ &\quad - \left[E_{t+1} \left(\sum_{j=0}^{\infty} \rho^j r_{t+1+j} \right) - E_t \left(\sum_{j=0}^{\infty} \rho^j r_{t+1+j} \right) \right]. \end{aligned}$$

With $E_t(r_{t+1+j}) = \bar{r} + \phi^j x_t$ and $x_{t+1} = \phi x_t + \varepsilon_{t+1}$, the revision in expected future returns equals $\frac{\rho \varepsilon_{t+1}}{1-\rho\phi}$. Hence:

Return decomposition:

$$r_{t+1} = \bar{r} + x_t + \eta_{d,t+1} - \frac{\rho \varepsilon_{t+1}}{1-\rho\phi},$$

where $\eta_{d,t+1}$ is the news about future dividend growth.

Variance of returns. From the AR(1) $x_t = \phi x_{t-1} + \varepsilon_t$, we have $\sigma_x^2 = \sigma_\varepsilon^2 / (1 - \phi^2)$, so $\sigma_\varepsilon^2 = (1 - \phi^2) \sigma_x^2$. Substituting:

$$\text{Var}(r_{t+1}) = \sigma_d^2 + \frac{1 + \rho^2 - 2\rho\phi}{(1 - \rho\phi)^2} \sigma_x^2 \xrightarrow{\rho \rightarrow 1} \sigma_d^2 + \frac{2\sigma_x^2}{1 - \phi}.$$

When $\phi \approx 1$, return variance is enormous — consistent with observed stock market volatility being too high relative to dividend growth alone (Shiller's excess volatility puzzle).

Long-horizon R^2 :

$$\frac{R^2(k)}{R^2(1)} = \left(\frac{1 - \phi^k}{1 - \phi} \right)^2 \cdot \frac{1}{VR(k)} \cdot \frac{1}{k}.$$

For small k : $\approx k/VR(k)$ (increasing); for large k with fixed ϕ : decreasing. Hence $R^2(k)$ is hump-shaped.

Time Correlation Problem

For $k = 2$ under $H_0 : \beta_1(2) = 0$, consecutive observations share one period:

$$\begin{aligned} r_1 + r_2 &= \beta_0(2) + u_{1,2} = 2\beta_0(1) + \varepsilon_1 + \varepsilon_2, \\ r_2 + r_3 &= \beta_0(2) + u_{2,2} = 2\beta_0(1) + \varepsilon_2 + \varepsilon_3, \\ r_3 + r_4 &= \beta_0(2) + u_{3,2} = 2\beta_0(1) + \varepsilon_3 + \varepsilon_4. \end{aligned}$$

The error terms share ε_2 between observations 1 and 2, ε_3 between observations 2 and 3, etc. Overlapping k -period regressions therefore induce serial correlation in $u_{t,t+k}$ up to lag $k - 1$:

$$\text{Cov}(u_{i,k}, u_{j,k}) \neq 0 \quad \forall |i - j| < k.$$

Newey–West Standard Errors

For $r_{t+1} + \dots + r_{t+k} = \beta_0 + \beta_1 x_t + u_{t,t+k}$,

$$\sqrt{T}(b_1 - \beta_1) = \left(\frac{1}{T} \sum x_t x_t' \right)^{-1} \frac{1}{\sqrt{T}} \sum x_t u_t \equiv Q^{-1} \frac{1}{\sqrt{T}} \sum Z_t.$$

Asymptotic variance: $\sqrt{T}(b_1 - \beta_1) \sim N(0, Q^{-1} S Q^{-1})$, where $S = \lim_{T \rightarrow \infty} T \cdot E(\bar{Z}_T \bar{Z}_T')$.

To unpack S , let $\Gamma_j = E[(y_t - \mu)(y_{t-j} - \mu)']$ denote the j -th autocovariance. The sample average $\bar{y}_T$ has covariance:

$$T \cdot E[(\bar{y}_T - \mu)(\bar{y}_T - \mu)'] = \Gamma_0 + \frac{T-1}{T} \Gamma_1 + \dots + \frac{1}{T} \Gamma_{T-1} + \frac{T-1}{T} \Gamma_{-1} + \dots + \frac{1}{T} \Gamma_{-(T-1)}.$$

For $y_t \sim MA(q)$: $S = \sum_{v=-q}^q \Gamma_v$. Sample estimate:

$$\hat{\Gamma}_v = \frac{1}{T} \sum_{t=v+1}^T (y_t - \bar{y})(y_{t-v} - \bar{y})', \quad \hat{S} = \hat{\Gamma}_0 + \sum_{v=1}^q (\hat{\Gamma}_v + \hat{\Gamma}_v').$$

Newey–West estimator (uses Bartlett kernel to guarantee positive semi-definiteness):

$$\tilde{S} = \hat{\Gamma}_0 + \sum_{v=1}^q \left(1 - \frac{v}{q+1} \right) (\hat{\Gamma}_v + \hat{\Gamma}_v'), \quad \hat{\Gamma}_v = \frac{1}{T} \sum_{t=v+1}^T (y_t - \bar{y})(y_{t-v} - \bar{y})'.$$

$$\sqrt{T}(\hat{\beta}_1(k) - \beta_1(k)) \sim N(0, Q^{-1} \tilde{S} Q^{-1}).$$

Spurious Regression (Stambaugh 1999)

The D/P ratio is persistent (near $I(1)$), and when summed, the cumulative return is also $I(1)$. Regressing one $I(1)$ series on another $I(1)$ series risks **spurious regression**. Stambaugh (1999) shows this also introduces bias in the single-period case because P_t drives both r_t and $d_t - p_t$, making their error terms correlated.

Bootstrap Correction

1. Estimate under H_0 to get $\hat{\beta}_0, \hat{\rho}_1$, residuals $\{\hat{\varepsilon}_t\}, \{\hat{u}_t\}$.
2. Resample residuals with replacement to get $\{\tilde{\varepsilon}_t\}, \{\tilde{u}_t\}$.
3. Generate bootstrap data:

$$\tilde{r}_{t+1}^{(j)} = \hat{\beta}_0 + \tilde{\varepsilon}_{t+1}^{(j)}, \quad (\widetilde{d_{t+1} - p_{t+1}})^{(j)} = \hat{\rho}_0 + \hat{\rho}_1 (\widetilde{d_t - p_t})^{(j)} + \tilde{u}_{t+1}^{(j)}.$$

4. Re-estimate β_1 for each bootstrap replication $j = 1, \dots, B$. Use the empirical distribution to correct for bias.

Exchange Rate

Interest Rate Parity

From the Korean (domestic) investor's perspective:

$$\begin{aligned}\text{UIP: } 1 + i_t &= E_t \left[\frac{(1 + i_t^*) S_{t+1}}{S_t} \right] \\ \text{CIP: } 1 + i_t &= \frac{(1 + i_t^*) F_t}{S_t}\end{aligned}$$

From the US (foreign) investor's perspective:

$$\begin{aligned}\text{UIP: } 1 + i_t^* &= E_t \left[\frac{S_t(1 + i_t)}{S_{t+1}} \right] \\ \text{CIP: } 1 + i_t^* &= \frac{S_t(1 + i_t)}{F_t}\end{aligned}$$

Under risk-neutral rational expectations, UIP and CIP imply $E_t(S_{t+1}) = F_t$.

(S_t : exchange rate KRW/USD; i_t : Korean rate; i_t^* : US rate; F_t : forward rate.)

Jensen's inequality. Since $f(x) = 1/x$ is convex:

$$E_t \left(\frac{1}{S_{t+1}} \right) > \frac{1}{E_t(S_{t+1})},$$

so $F_t = E_t(S_{t+1})$ does *not* imply $1/F_t = E_t(1/S_{t+1})$, meaning the two parity conditions from the two perspectives are not simultaneously consistent:

$$F_t = E_t(S_{t+1}) \not\Rightarrow F_t \neq E_t(S_{t+1}).$$

Log-normality resolution (risk-neutral case). Using $\ln E[e^X] = E[X] + \frac{1}{2} \text{Var}(X)$:

$$\text{In Korea: } f_t = \ln E_t(e^{s_{t+1}}) = E_t(s_{t+1}) + \frac{1}{2} \sigma_t^2(s_{t+1}).$$

$$\text{In US: } -f_t = -\ln E_t(e^{s_{t+1}}) = -E_t(s_{t+1}) + \frac{1}{2} \sigma_t^2(s_{t+1}).$$

The two views are reconciled by introducing price-level risk. When UIP and CIP hold simultaneously with real price deflation, equating $E_t \left(\frac{F_t}{P_{t+1}} \right) = E_t \left(\frac{S_{t+1}}{P_{t+1}} \right)$ and applying log-normality yields:

$$f_t = E_t(s_{t+1}) + \frac{1}{2} \sigma_t^2(s_{t+1}) - \text{Cov}_t(s_{t+1}, p_{t+1}).$$

Regression Models and the UIP Puzzle

1. $s_{t+1} - s_t = \beta_0 + \beta_1(i_t - i_t^*) + u_{t+1}$
2. $s_{t+1} - s_t = \beta_0 + (f_t - s_t) + u_{t+1}$

Theory predicts $\beta_1 = 1$. Empirically, $\hat{\beta}_1$ is often **significantly negative** — the **UIP puzzle**.

Derivation: Interpreting the UIP coefficient

$$\beta_1 = \frac{\text{Cov}(f_t - s_t, s_{t+1} - s_t)}{\text{Var}(f_t - s_t)} = \frac{\text{Cov}(f_t - s_t, E_t(s_{t+1}) - s_t)}{\text{Var}(f_t - s_t)}.$$

Define forward premium risk premium $rp_f = (f_t - s_t) - (E_t(s_{t+1}) - s_t)$. Then:

$$\text{Var}(rp_f) = (1 - 2\hat{\beta}_1)\text{Var}(f_t - s_t) + \text{Var}(E_t(s_{t+1}) - s_t).$$

If $\hat{\beta}_1 < \frac{1}{2}$, the risk premium is *more volatile* than the expected exchange rate change — consistent with a significantly negative estimated coefficient.

Exchange Rate Predictability

Meese & Rogoff (1983) showed structural models fail to out-predict the random walk out-of-sample.

Flexible Price Monetary Model

Four building blocks (domestic and foreign):

$$\begin{aligned} E_t(s_{t+1}) - s_t &= \delta(i_t - i_t^*) && \text{(UIP)} \\ s_t &= \beta(p_t - p_t^*) && \text{(PPP)} \\ m_t &= p_t - \phi i_t + \gamma y_t && \text{(Money demand, home)} \\ m_t^* &= p_t^* - \phi i_t^* + \gamma y_t^* && \text{(Money demand, foreign)} \end{aligned}$$

Subtracting the foreign from the domestic money demand equation:

$$m_t - m_t^* = (p_t - p_t^*) - \phi(i_t - i_t^*) + \gamma(y_t - y_t^*).$$

Substituting PPP ($p_t - p_t^* = s_t/\beta$) and UIP ($i_t - i_t^* = (E_t(s_{t+1}) - s_t)/\delta$):

$$m_t - m_t^* = \frac{s_t}{\beta} - \frac{\phi}{\delta}(E_t(s_{t+1}) - s_t) + \gamma(y_t - y_t^*).$$

Solving for s_t :

$$\left(\frac{1}{\beta} + \frac{\phi}{\delta}\right) s_t = (m_t - m_t^*) - \gamma(y_t - y_t^*) + \frac{\phi}{\delta} E_t(s_{t+1}).$$

Combining:

$$s_t = \lambda_1(m_t - m_t^*) - \gamma\lambda_1(y_t - y_t^*) + \lambda_2 E_t(s_{t+1}),$$

where $\lambda_1 = \frac{\beta\phi}{\delta + \beta\phi}$ and $\lambda_2 = \frac{\beta\phi}{\delta + \beta\phi}$.

Iterating forward (no-bubble):

$$s_t = \lambda_1 E_t \left[\sum_{i=0}^{\infty} \lambda_2^i (m_{t+i} - m_{t+i}^*) \right] - \gamma\lambda_1 E_t \left[\sum_{i=0}^{\infty} \lambda_2^i (y_{t+i} - y_{t+i}^*) \right].$$

With $\delta = \beta = 1$ (so $\lambda_1 = \lambda_2 = \phi/(1 + \phi)$ and $\lambda_1 + \lambda_2 = 1$), define the fundamental $f_t = (m_t - m_t^*) - \gamma(y_t - y_t^*)$. Then:

Derivation: Derivation of $s_t - f_t$ cointegration

$$\begin{aligned} s_t &= (1 - \lambda_2) E_t \left[\sum_{i=0}^{\infty} \lambda_2^i f_{t+i} \right] \\ &= E_t[(1 - \lambda_2) f_t + (1 - \lambda_2) \lambda_2 f_{t+1} + \dots] \\ &= f_t - \lambda_2 f_t + (1 - \lambda_2) \lambda_2 E_t(f_{t+1}) + \dots \end{aligned}$$

Subtracting f_t :

$$\begin{aligned} s_t - f_t &= -\lambda_2 f_t + \lambda_2 E_t(f_{t+1}) - \lambda_2^2 E_t(f_{t+1}) + \lambda_2^2 E_t(f_{t+2}) - \dots \\ &= \lambda_2 (E_t(f_{t+1}) - f_t) + \lambda_2^2 (E_t(f_{t+2}) - E_t(f_{t+1})) + \dots \\ &= E_t[\lambda_2 \Delta f_{t+1} + \lambda_2^2 \Delta f_{t+2} + \lambda_2^3 \Delta f_{t+3} + \dots]. \end{aligned}$$

$$s_t - f_t = E_t \left[\sum_{i=1}^{\infty} \lambda_2^i \Delta f_{t+i} \right].$$

Since Δf is $I(0)$ and $0 < \lambda_2 < 1$, the right side is $I(0)$: s_t and f_t are **cointegrated** with vector $[1, -1]$.

Taylor Rule Model

Central banks set rates according to:

$$\begin{aligned} i_t &= r_g(q_t - c_0) + r_\pi E_t(\pi_{t+1}) + r_g y_t && \text{(Korea)} \\ i_t^* &= r_\pi E_t(\pi_{t+1}^*) + r_g y_t^* && \text{(Foreign)} \end{aligned}$$

Since $s_t = q_t + p_t - p_t^*$ (nominal = real exchange rate + price differential), UIP $i_t - i_t^* = E_t(s_{t+1} - s_t)$ becomes:

$$i_t - i_t^* = E_t(q_{t+1}) - q_t + E_t(\pi_{t+1} - \pi_{t+1}^*).$$

Subtracting the two Taylor rules (Korea – Foreign):

$$i_t - i_t^* = r_g(q_t - c_0) + r_\pi E_t(\pi_{t+1} - \pi_{t+1}^*) + r_g(y_t - y_t^*).$$

Equating the two expressions for $i_t - i_t^*$:

$$r_g(q_t - c_0) + r_\pi E_t(\pi_{t+1} - \pi_{t+1}^*) + r_g(y_t - y_t^*) = E_t(q_{t+1}) - q_t + E_t(\pi_{t+1} - \pi_{t+1}^*).$$

Rearranging $(1 + r_g)q_t$:

Differencing and using UIP leads to a forward-looking equation for the real exchange rate:

$$q_t = b r_g c_0 + b(1 - r_\pi) E_t(\pi_{t+1} - \pi_{t+1}^*) - b r_g(y_t - y_t^*) + b E_t(q_{t+1}),$$

where $b = 1/(1 + r_g)$. Iterating forward:

$$q_t = \frac{b r_g c_0}{1 - b} + b(1 - r_\pi) E_t \left[\sum_{i=0}^{\infty} b^i (\pi_{t+1+i} - \pi_{t+1+i}^*) \right] - b r_g E_t \left[\sum_{i=0}^{\infty} b^i (y_{t+1+i} - y_{t+1+i}^*) \right],$$

under $\lim_{j \rightarrow \infty} b^j E_t(q_{t+j}) = 0$.

End of Notes — Last compiled June 30, 2026